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ORIGINAL ARTICLE

Utilization of Serum Metabolomics and Polygenic Risk Scores in a Novel Risk Stratification Tool for the Prediction of Incident Atrial Fibrillation

Subhanik Purkayastha, MD; Joseph Park, MD, PhD; Sebastian Beyer, MD; Arun Chandra, MD; Steven M. Markowitz, MD; Bruce B. Lerman, MD; Olivier Elemento, PhD; Jan Krumsiek, PhD; James C. Lo, MD; Jim W. Cheung, MD

BACKGROUND: Atrial fibrillation (AF) is associated with substantial morbidity and mortality. We sought to investigate the predictive value of serum metabolomics for 5-year incident AF in the context of clinical and polygenic risk score (PRS) stratification tools.

METHODS: We studied a cohort of 240 628 patients UK Biobank participants with proton nuclear magnetic resonance spectroscopy measurements of 170 serum metabolites at enrollment. Five-year incidence of AF was assessed using Cox proportional hazards models. Cohorts for Heart and Aging Research in Genomic Epidemiology-AF and AF polygenic risk score (AF-PRS) scores were used as benchmark risk models for comparison. Models were trained on 80% of the cohort, and performances were validated on the remaining 20% cohort. Performance of clinical, AF-PRS, and combined metabolomics models was evaluated using time-dependent area under the receiver operating characteristic curve, net reclassification improvement, and relative integrated discrimination improvement analysis.

RESULTS: During follow-up, 4174 (1.7%) participants developed AF. After training a model on the full metabolomics panel in addition to Cohorts for Heart and Aging Research in Genomic Epidemiology-AF and AF-PRS, the final model retained 8 metabolites. Creatinine level was associated with increased risk (hazard ratio, 1.01 per 1 SD log-transformed value [95% CI, 1.00–1.03]) while linoleic acid level (hazard ratio, 0.985 [0.979–0.994]) was associated with decreased risk of AF. The addition of metabolomics to the Cohorts for Heart and Aging Research in Genomic Epidemiology-AF+AF-PRS model improved risk prediction (5-year time-dependent area under the receiver operating characteristic curve, 0.789 [0.776–0.802] versus 0.755 [0.738–0.772]; $P<0.05$) and stratification on the validation set (NRI_{cases}: 11.1%, $NRI_{controls}$: 3.1%, $IDI_{relative}$: 11.6%). A model using only age, sex, metabolomics, and AF-PRS had fair risk prediction on the validation set (5-year time-dependent area under the receiver operating characteristic curve, 0.787 [0.773–0.801]).

CONCLUSIONS: The addition of metabolomics to clinical and genomic risk scores improves the prediction of 5-year incident AF. A risk stratification tool using age, sex, and serum metabolomics and AF-PRS provides excellent AF risk prediction. Mechanisms by which specific metabolites reflect AF risk require further exploration.

Key Words: atrial fibrillation ■ heart failure ■ incidence ■ ischemic stroke ■ quality of life

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Atrial fibrillation (AF) is associated with substantial morbidity and mortality globally. As the most commonly encountered sustained arrhythmia, AF can significantly impact a patient's quality of life and may be associated with severe complications such as stroke, heart failure, and mortality.[1] Furthermore, since AF is often paroxysmal and can be asymptomatic, ischemic stroke can be the first manifestation of undetected AF.[2]

Until recently, risk prediction for AF has mainly relied on the use of clinical variables. Clinical risk scores, such as the Cohorts for Heart and Aging Research in Genomic Epidemiology-AF (CHARGE-AF) and Electronic Health Record-AF, use variables derived from patient medical

Correspondence to: Jim W. Cheung, MD, Division of Cardiology, Weill Cornell Medicine, 520 E 70th St, 4th Floor, New York, NY 10021. Email jac9029@med.cornell.edu
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WHAT IS KNOWN?

- The utility of metabolomics for the prediction of atrial fibrillation risk in the general population has not been well-established.

WHAT THE STUDY ADDS

- In this population-based study of the UK Biobank involving >240 000 participants, we demonstrate that the addition of metabolomics to benchmark clinical risk and polygenic risk models results in significantly improved atrial fibrillation risk prediction.
- A risk stratification tool using age, sex, serum metabolomics, and atrial fibrillation polygenic risk score offers excellent atrial fibrillation risk prediction.

Nonstandard Abbreviations and Acronyms

AF	atrial fibrillation
AF-PRS	atrial fibrillation polygenic risk score
ARIC	Atherosclerosis Risk in Communities
BNP	B-type natriuretic peptide
CHARGE-AF	Cohorts for Heart and Aging Research in Genomic Epidemiology-AF
CHS	Cardiovascular Health Study
FHS	Framingham Heart Study
HR	hazard ratio
hs-CRP	high-sensitivity C-reactive protein
ICD	*International Classification of Diseases*
NRI	net reclassification improvement
td-AUC	time-dependent area under the receiver operating curve
UKBB	UK Biobank

histories to estimate the risk of developing AF within 5 years.[3,4] Recently, models such as Predict-AF have incorporated polygenic risk scores (PRS) to account for predetermined genetic risk associated with AF.[5] It has been shown that PRS is an independent risk factor of AF and complements clinical risk scores in predictive value.[6,7]

Although genomic data offers significant value in stratifying AF risk, it is nonmodifiable and does not account for environmental factors and disease progression.[8] Metabolomics captures the real-time state of cellular processes. Comprehensive measurement of small molecule metabolites in biological samples can be performed using proton nuclear magnetic resonance spectroscopy. Given that proton nuclear magnetic resonance has minimal batch effects, progressively lower cost, and high throughput, its use in exploring the human blood metabolome has gained increasing interest in recent years.[9] It has been reported that inflammation, oxidative stress, and cellular metabolism are involved in AF pathogenesis, which motivates the use of metabolomics to better understand risk profiles.[10,11] Studies conducted on small cohorts have identified specific metabolites associated with AF pathogenesis, ranging from small peptides to microRNAs to end-products of metabolism.[12–14] Metabolomic profiles in AF have been shown to correlate with clinical status and predict AF sequela.[15,16] Although these studies have demonstrated the utility of metabolomics in AF risk stratification, its additional predictive value over established risk stratification tools using clinical and PRS is unknown. We hypothesize that metabolomics will capture additional AF risk independently of clinical and genetic factors and that a prediction model utilizing age, sex, PRS, and serum metabolome alone would have comparable or improved performance in predicting 5-year AF risk compared with clinical risk scores alone.

Using the UK Biobank (UKBB) cohort, this study aims to (1) develop elastic net-regularized Cox proportional hazards models using a training set containing 80% of the study cohort to generate scores based on varying combinations of age, sex, CHARGE-AF score, AF-PRS, and metabolites, (2) assess and compare predictive performance of these models on a validation set containing 20% of the study cohort, and (3) identify individual metabolites associated with 5-year incident AF.

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METHODS

Study Database

Established in 2006, the UKBB is a large-scale prospective cohort that voluntarily enrolled ≈500 000 individuals who were aged 40 to 69 years, showed capacity to consent, and lived within 25 miles of one of the assessment centers. Among 22 different assessment centers spread across the United Kingdom, participants received extensive baseline characterization via touchscreen questionnaires, verbal interviews, physical measurements, and the collection of relevant biological specimens.[17] All participants gave written, informed consent. UKBB continues to update the database with new imaging, biomarker, and clinical data. The North West Multi-center Research Ethics Committee has approved the use of UKBB data by all researchers who submit successful applications. For this study, data were accessed under application ID 137851. This study was deemed exempt by the Weill Cornell institutional review board as it uses a deidentified patient database. The data that support the findings of this study are available from the corresponding author on reasonable request. The research reported in this article adhered to the revised 2013 Helsinki Declaration guidelines.

Study Population

We included all UKBB participants with nonmissing values for the complete panel of 170 original proton nuclear magnetic resonance metabolite measurements taken from their initial assessment center visit blood draw (n=256 550). We excluded all participants with incomplete AF-PRS (n=2489)

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and CHARGE-AF score (n=13 433) variable components. To assess the incidence of AF, we excluded all patients with prevalent AF (n=4543). Prevalent AF was defined using 2 methods: (1) self-report of previously diagnosed AF or atrial flutter during an interview with a trained nurse at baseline enrollment visit (n=2020) and (2) identification of *International Classification of Diseases, Ninth and Tenth Revisions* (*ICD-9* and *ICD-10*) coded diagnosis of AF or atrial flutter dated before the date of enrollment visit (n=2523). The final study cohort consisted of 240 628 patients (Figure 1).

End Point Definitions

The study end point was the incidence of AF within 5 years of baseline enrollment. The 5-year end point was chosen to permit direct comparisons to the clinical CHARGE-AF score, which was validated for 5-year AF prediction. For this study, both AF and atrial flutter were included in the incident AF end point. Disease phenotypes were identified using self-reported histories and *ICD-9* and *ICD-10* codes found in primary care, hospitalization, surgical, procedural, and death registry records. Incident AF was defined as the presence of a relevant ICD-9 or ICD-10 code during longitudinal follow-up. We used data fields defined in 2 prior UKBB AF studies.[18,19] The full list of AF data fields used in this study is shown in Table S1. Participants without prevalent AF were followed from the date of their initial assessment center visit until the date of first AF occurrence, death, loss to follow-up, or end of the 5-year follow-up, whichever occurred first.

Benchmark Clinical and PRS

The benchmark clinical model we used was derived from the CHARGE-AF.[3] This model was derived from 3 large US-based cohorts (ARIC study [Atherosclerosis Risk in Communities], the CHS [Cardiovascular Health Study], and the FHS [Framingham Heart Study]) and validated in 2 separate cohorts from the Netherlands (Rotterdam Study) and Iceland (Age, Gene and Environment—Reykjavik study). It is a model that incorporates age, race, height, weight, systolic and diastolic blood pressure, smoking status, antihypertensive medication use, and histories of diabetes, heart failure, and myocardial infarction. The 5-year CHARGE-AF score is computed with the following formula: $1-0.9718412736^{(\Sigma\beta X-12.58156)}$, where β is the regression coefficient, and X is the value or level for each risk factor. Final CHARGE-AF scores were log-normalized and standardized. The clinical variables in CHARGE-AF were obtained from the UKBB using the codes found in Table S2. The values for each

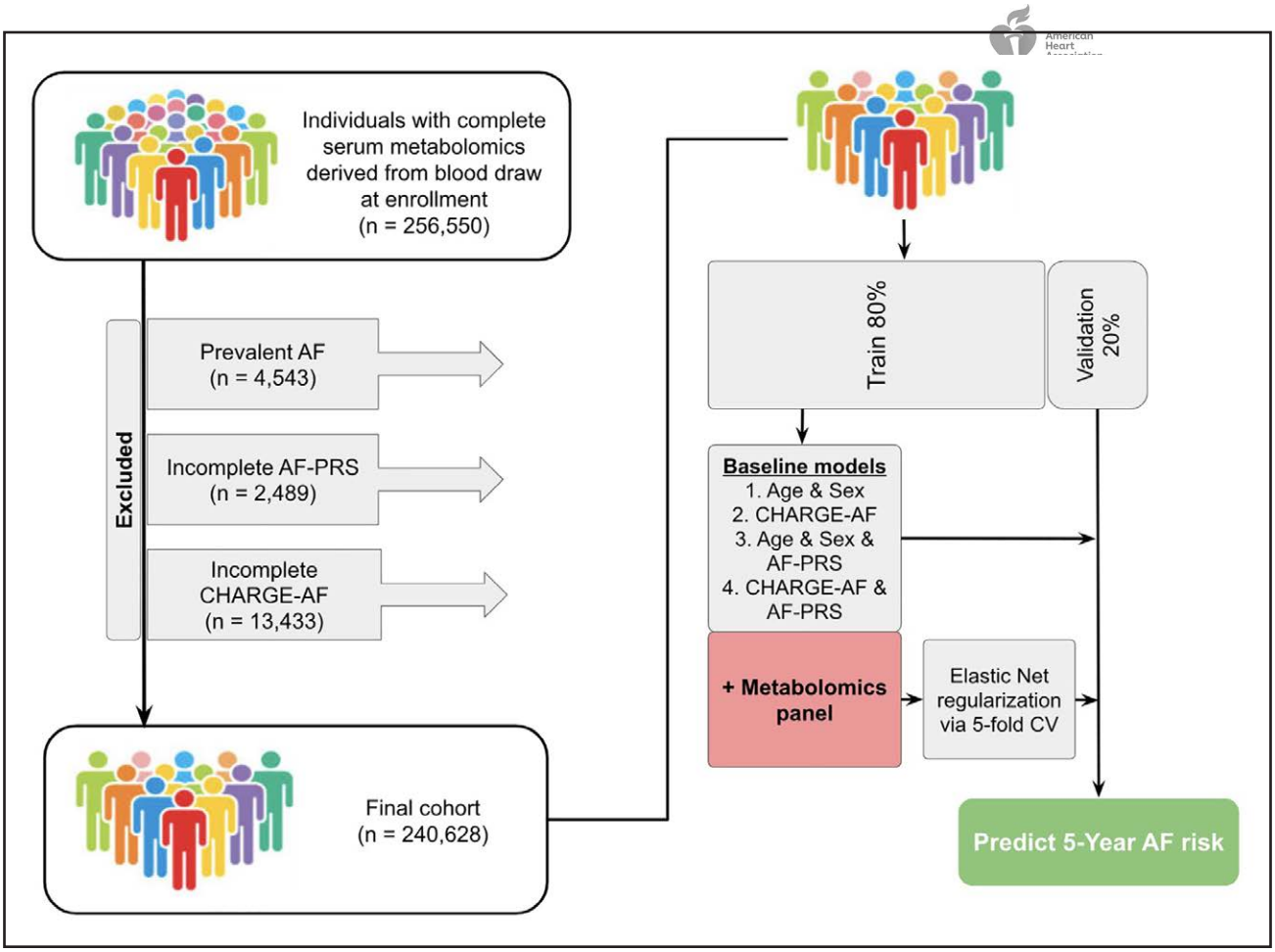

Figure 1. Study the flowchart diagram.
Schematic showing the study population after inclusion and exclusion criteria, with subsequent risk prediction workflow. AF indicates atrial fibrillation; AF-PRS, AF-polygenic risk score; CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology–Atrial Fibrillation; and CV, cross-validation.

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covariate can be found in Table S3. The benchmark PRS for AF was obtained by the UKBB PRS Release, which consisted of well-powered PRSs for 28 diseases and 25 quantitative traits.[20] This release consisted of 2 PRS sets, a standard set, which was trained on external (non-UKBB) data only, and an Enhanced set, which was trained on both external and internal data. The standard set, which was used for the current study, incorporates ancestry adjustment through a principal component-based centering step and standardization within ancestry groups, ensuring that the score distributions are approximately comparable across populations.

Metabolomics Measurement

Non-fasting EDTA plasma samples were drawn for every UKBB participant at their initial baseline enrollment visit. Metabolite measurements were recorded from this initial serum sample. As of July 2023, Nightingale Health has conducted high-throughput proton nuclear magnetic resonance spectroscopy on ≈275 000 patients' serum samples.[21] Further details regarding the spectroscopy and quality control processes can be found here: https://biobank.ndph.ox.ac.uk/showcase/refer.cgi?id=130. Although the initial metabolomics panel consisted of 251 measurements, we limited our analyses to the 170 absolute measures (concentrations measured in millimoles per liter and diameters measured in nanometers) and removed 81 ratio measures. Metabolites were manually grouped according to relevant pathways. The full list of metabolite groups and individual metabolites can be found in Table S4.

Metabolomics Model Derivation and Validation

The entire cohort was randomized into 80/20 derivation/validation cohorts with roughly equal proportions of AF cases to controls. Baseline CPH models without metabolomics were trained using 4 combinations of variables: (1) age and sex; (2) age, sex, and AF-PRS; (3) CHARGE-AF; and (4) CHARGE-AF and AF-PRS. Then, we trained the same 4 CPH models, but this time with the addition of the full panel of 170 metabolites, with an additional CPH model trained only on the metabolomics panel. Hence, a total of 9 models were trained and tested: (1) age and sex; (2) metabolomics only; (3) age and sex+metabolomics; (4) age, sex, and AF-PRS; (5) age, sex, and AF-PRS+metabolomics; (6) CHARGE-AF; (7) CHARGE-AF+metabolomics; (8) CHARGE-AF and AF-PRS; and (9) CHARGE-AF and AF-PRS+metabolomics. All metabolite measurements were log-normalized and standardized. The following methods were adapted from prior work done on metabolomic risk prediction in heart failure by Oexner et al.[22] For models that included metabolomics, we used elastic net regularization as our feature reduction technique because (1) it allows correlated metabolites to be retained together with appropriately shrunk coefficients, thereby preserving metabolic pathway-level information and (2) it yields interpretable coefficients for individual metabolites. We performed 5-fold cross-validation to optimize the hyperparameters of lambda, which controls the overall regularization strength, and alpha, which controls the balance between the Lasso and Ridge penalties. After training all 8 models, we evaluated each of their performances on the validation set. A study workflow schematic is shown in Figure 1.

Individual Metabolite Associations

To further understand which metabolites modulate AF risk, CPH models were fitted using each of the 170 metabolite measurements adjusted for one of age and sex, CHARGE-AF, and AF-PRS. Before fitting the CPH models, all metabolite measurements were log-normalized and standardized. We obtained hazard ratios (HRs) of 5-year incident AF for each individual metabolite.

Statistical Analysis

All analyses were conducted using Python 3.7 within the Jupyter notebook environment on the UKBB Research Analysis Platform. Statistical tests performed on the baseline characteristics were the Mann-Whitney *U* test for continuous variables and the χ^2 test for categorical variables, using the scipy.stats library. Cox proportional hazards models were implemented using the scikit-survival library, specifically CoxPHSurvivalAnalysis and CoxnetSurvivalAnalysis. Covariate-level proportional hazards were assessed by testing the correlation between each covariate and event times among uncensored events using Spearman correlation. Covariates with $P>0.05$ were considered to satisfy the proportional hazards assumption. Model performance and covariate effect sizes were evaluated using 1000 bootstrap resamples to estimate 95% CIs for time-dependent area under the receiver operating curve (td-AUC), Harrell C-index, sensitivity, specificity, net reclassification improvement (NRI), and relative integrated discrimination improvement. Differences between models were considered statistically significant if the 95% CI of the difference excluded zero. Sensitivity and specificity in the validation set were computed at thresholds maximizing the Youden index, including only individuals with full 5-year follow-up. A significance threshold of $P<0.05$ was applied for all tests, and Bonferroni correction was used for analyses involving multiple comparisons (eg, individual metabolite associations). Complete code is available on request.

RESULTS

In the study cohort of 240 628 individuals, the mean age was 56 (interquartile range, 50–62), and there were 110 022 (46%) males. During median follow-up of 5 years (≈1.2 million person-years), 4174 (1.7%) patients developed AF. We observed significant differences in baseline characteristics among healthy controls and those who were diagnosed with AF within 5 years. Individuals with AF were more likely to be older, male, and have a higher comorbidity burden of hypertension, diabetes, heart failure, and coronary artery disease (Table 1). Compared with individuals without AF at 5 years, individuals with incident AF had significantly higher AF-PRS (0.58 versus 0.088; $P<0.001$) and CHARGE-AF scores (0.022 versus 0.011; $P<0.001$).

Metabolomics and AF Risk Prediction

The entire cohort was split into 80:20 derivation:validation cohorts, with each partition being well-balanced in terms of AF incidence. The baseline

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Table 1. Study Population Baseline Characteristics

Characteristic	Entire cohort, n=240 628	5-year incident atrial fibrillation		*P* value
		Yes (n=4174)	No (n=236 454)	
Age at recruitment, y, mean (IQR)	56 (50–62)	62 (60–67)	56 (50–63)	<0.001
Male, n (%)	110 022 (46%)	2707 (65%)	107 315 (45%)	<0.001
White, n (%)	240 372 (>99%)	4161 (>99%)	236 211 (>99%)	<0.001
Physical measurements, mean (IQR)				
Height, cm	168 (162–175)	171 (165–178)	168 (162–175)	<0.001
Weight, kg	78 (67–87)	85 (73–95)	78 (66–87)	<0.001
SBP, mm Hg	134 (121–145)	139 (126–151)	134 (121–145)	<0.001
DBP, mm Hg	80 (73–87)	81 (73–88)	80 (73–87)	0.006
Comorbidities, n (%)				
Hypertension	65 959 (27%)	2125 (51%)	63 834 (27%)	<0.001
Diabetes	12 536 (5%)	447 (11%)	12 089 (5%)	<0.001
Heart failure	227 (0.09%)	24 (0.6%)	203 (0.09%)	<0.001
Myocardial infarction	5376 (2%)	359 (9%)	5017 (2%)	<0.001
Smoking status (current)	25 058 (10%)	465 (11%)	24 593 (10%)	0.127
Scores, mean (IQR)				
AF-PRS	0.097 (−0.52 to 0.70)	0.58 (−0.08 to 1.19)	0.088 (−0.53 to 0.69)	<0.001
CHARGE-AF	0.011 (0.0033–0.015)	0.022 (0.011–0.029)	0.011 (0.0033–0.015)	<0.001

AF-PRS indicates AF polygenic risk score; CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology–Atrial Fibrillation; DBP, diastolic blood pressure; IQR, interquartile range; and SBP, systolic blood pressure.

characteristics of the derivation and validation cohorts are shown in Table S5. After training the models using an elastic net, varying numbers of metabolites were retained in the final models. For the metabolomics-only model, 51 of 170 metabolites were included (30.0%). For the age and sex+metabolomics model, 118 of 170 metabolites (69.4%) were included. For the age and sex+AF-PRS+metabolomics model, 112 of 170 metabolites (65.9%) were included. For both the CHARGE-AF+metabolomics and the CHARGE-AF+AF-PRS+metabolomics models, only 8 of 170 metabolites (4.7%) were included in the 2 models. For the CHARGE-AF+AF-PRS+metabolomics model and the age and sex+AF-PRS+metabolomics model, the specific metabolites and their associated HRs are shown in Figure S1A and S1B, respectively.

Trained models were then evaluated on the validation set. To isolate the predictive value of metabolomics independently, the metabolomics-only model had a 5-year td-AUC of 0.680 (95% CI, 0.659–0.699). The addition of metabolomics improved the predictive accuracy and discrimination of all clinical AF risk prediction models, as reflected in the changes in 5-year td-AUC, C-index, NRI, and integrated discrimination improvement shown in Tables 2 and 3. For the age and sex model, the inclusion of metabolomics resulted in a Δtd-AUC=+0.024. This was associated with an overall NRI of 20.5% and a relative integrated discrimination improvement of 14.5%. The model trained on age and sex+metabolomics performed nearly as well as the model trained only on CHARGE-AF (td-AUC, 0.761 versus 0.763).

Furthermore, the addition of metabolomics increased the predictive accuracy of models that incorporated clinical and polygenic risk. A model combining age, sex, and AF-PRS exhibited a notable enhancement on inclusion of metabolomics with a Δtd-AUC=+0.019. In addition, significant improvement was observed in the CHARGE-AF+AF-PRS model, where the td-AUC increased from 0.755 (95% CI, 0.738–0.772) to 0.786 (95% CI, 0.776–0.802) on the addition of metabolomics, resulting in the highest Δtd-AUC=+0.034. The addition of metabolomics to CHARGE-AF and AF-PRS resulted in an overall NRI of 14.2% and a substantial relative integrated discrimination improvement of 11.6%. Time-dependent AUCs up to 5 years for each model are shown in Table 4. Overall, the models with the highest AF predictive accuracy were the age and sex+AF-PRS+metabolomics model and the CHARGE-AF+AF-PRS+metabolomics model. On the validation set, both models had nearly identical td-AUC (0.784 versus 0.786). Furthermore, the age and sex+AF-PRS+metabolomics model correctly reclassified AF cases to higher risk more often than the CHARGE-AF+AF-PRS+metabolomics model (NRI_{cases}=18.6%), although it reclassified controls to lower risk less often ($NRI_{controls}$=−10.2%).

Cumulative AF incidence at 5 risk quintiles was computed for all models on the validation set, as shown in Figure 2. At the highest risk quintile, cumulative AF incidence increased an average of 0.28% after adding metabolomics to the original models. At the lowest risk quintile, cumulative AF incidence decreased an average of 0.06% after adding metabolomics to the original

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Table 2. Five-Year Time-Dependent AUC, Harrel C-Index, Sensitivity, and Specificity Metrics of AF Risk Prediction Models on the Validation Set

Model	td-AUC @ 5 y	C-index	Sensitivity	Specificity
Age and sex	0.737 (0.722–0.754)	0.734 (0.717–0.749)	0.757 (0.677–0.800)	0.614 (0.594–0.693)
Metabolomics only	0.680 (0.659–0.699)	0.677 (0.658–0.698)	0.662 (0.628–0.697)	0.599 (0.594–0.603)
Age and sex+metabolomics	0.761 (0.745–0.776)	0.757 (0.741–0.771)	0.764 (0.675–0.819)	0.637 (0.591–0.725)
CHARGE-AF	0.763 (0.748–0.779)	0.760 (0.745–0.775)	0.794 (0.699–0.836)	0.619 (0.593–0.708)
CHARGE-AF+metabolomics	0.771 (0.755–0.785)	0.767 (0.753–0.782)	0.764 (0.711–0.850)	0.660 (0.571–0.702)
Age, sex, and AF-PRS	0.768 (0.753–0.783)	0.764 (0.749–0.779)	0.743 (0.695–0.779)	0.671 (0.652–0.695)
Age, sex, and AF-PRS+metabolomics	0.787 (0.773–0.801)	0.783 (0.769–0.798)	0.743 (0.648–0.839)	0.688 (0.589–0.774)
CHARGE-AF and AF-PRS	0.755 (0.738–0.772)	0.752 (0.736–0.769)	0.704 (0.568–0.808)	0.677 (0.564–0.814)
CHARGE-AF and AF-PRS+metabolomics	0.789 (0.776–0.802)	0.785 (0.771–0.798)	0.756 (0.716–0.834)	0.694 (0.610–0.724)

AF indicates atrial fibrillation; AF-PRS, AF-polygenic risk score; CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology—Atrial Fibrillation; and td-AUC, time-dependent area under the receiver operating curve.

models. Notably, the largest difference in cumulative AF incidence between the highest and lowest risk quintiles was seen for the age and sex+AF-PRS+metabolomics (Δincidence$_{\text{top quintile—bottom quintile}}$, 4.78%).

Individual Metabolite Associations With AF

On fitting CPH models for each individual metabolite, we observed significant HRs after adjusting for AF-PRS and either age and sex or CHARGE-AF. The HRs for individual metabolites adjusted by age, sex, and AF-PRS are shown in Figure 3, and the results from the other models are shown in Figure S2A through S2D. Out of 170 total metabolites, models adjusted for AF-PRS, age, and sex resulted in 132 significant associations (77.6%), whereas models adjusted by AF-PRS and CHARGE-AF resulted in 119 significant associations (70.0%). Metabolites involved in ketone body metabolism, including 3-hydroxybutyrate (HR per 1 SD log-transformed value, 1.06–1.10), acetoacetate (HR, 1.08–1.11), and acetone (HR, 1.08–1.11), were associated with a higher risk for AF for all models. Additionally, amino acids, such as tyrosine (HR, 1.08–1.16) and phenylalanine (HR, 1.06–1.08), along with creatinine (HR, 1.08–1.16), were associated with increased AF risk across all models. However, several metabolites were associated with decreased AF risk. These protective metabolites included apolipoproteins, fatty acids, and various lipid classes, including very low-density lipoprotein (LDL), LDL, and high-density lipoprotein species. One metabolite that was found to be the most protective from AF was linoleic acid (HR, 0.72–0.84), and this association was consistent across all models.

For models adjusted by patient's AF-PRS, age, and sex, the top 3 most significantly associated metabolites were linoleic acid (HR, 0.80 [95% CI, 0.78–0.82]; P=8.4×10^{-47}), cholesteryl esters in LDL (HR, 0.81 [95% CI, 0.78–0.83]; P=2.4×10^{-44}), and total cholesterol minus high-density lipoprotein cholesterol (HR, 0.81 [95% CI, 0.79–0.83]; P=1.4×10^{-43}). For models adjusted by AF-PRS and CHARGE-AF, the top 3 most significantly associated metabolites were linoleic acid (HR, 0.85 [95% CI, 0.82–0.88]; P=6.3×10^{-24}), albumin (HR, 0.87 [95% CI, 0.84–0.89]; P=3.4×10^{-20}), and omega-6 fatty acids (HR, 0.86 [95% CI, 0.84–0.89]; P=8.7×10^{-20}).

DISCUSSION

Our study demonstrates that incorporating serum metabolomics data into established clinical and genomic risk scores enhances the prediction of 5-year incident AF. The improvement in predictive performance was consistent

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Table 3. Net Reclassification Index and Integrated Discrimination Index Between Pairs of Models on the Validation Set

Metric	Age and sex vs age and sex+metabolomics	CHARGE-AF vs CHARGE-AF+metabolomics	Age, sex, and AF-PRS vs age, sex, and AF-PRS+metabolomics	CHARGE-AF and AF-PRS vs CHARGE-AF and AF-PRS+metabolomics	CHARGE-AF and AF-PRS+metabolomics vs age, sex, and AF-PRS+metabolomics
NRI					
Cases (%)	17.6 (10.9 to 23.8)	11.1 (4.2 to 18.2)	15.8 (9.5 to 22.6)	11.1 (4.1 to 17.8)	18.6 (11.7 to 25.9)
Controls (%)	2.9 (2.1 to 3.8)	1.5 (0.6 to 2.4)	3.0 (2.1 to 3.9)	3.1 (2.1 to 3.9)	−10.2 (−11.0 to −9.3)
IDI					
Relative (%)	14.5 (10.4 to 18.2)	2.7 (1.6 to 3.8)	10.2 (7.3 to 13.0)	11.6 (7.3 to 16.0)	−8.7 (−13.6 to −2.8)

AF-PRS indicates atrial fibrillation-polygenic risk score; CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology—Atrial Fibrillation; IDI, integrated discrimination improvement; and NRI, net reclassification improvement.

Table 4. Time-Dependent Area Under the Curve Evaluated at Each Year Up to 5 Years for All Models

Model	1 y	2 y	3 y	4 y	5 y
Age and sex	0.721	0.714	0.729	0.728	0.737
Metabolomics only	0.711	0.68	0.671	0.675	0.68
Age and sex+metabolomics	0.754	0.746	0.753	0.754	0.761
CHARGE-AF	0.738	0.74	0.755	0.76	0.764
CHARGE-AF+metabolomics	0.756	0.751	0.761	0.768	0.771
Age, sex, and AF-PRS	0.763	0.756	0.765	0.762	0.767
Age, sex, and PRS+metabolomics	0.786	0.778	0.783	0.782	0.787
CHARGE-AF and AF-PRS	0.752	0.751	0.759	0.755	0.755
CHARGE-AF and AF-PRS+metabolomics	0.778	0.773	0.781	0.786	0.788

AF-PRS indicates atrial fibrillation-polygenic risk score; and CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology—Atrial Fibrillation.

across all models, irrespective of the baseline approach used (age and sex, CHARGE-AF, and AF-PRS). Whenever metabolomics was added to a baseline model, 10% or more of AF cases were appropriately predicted to have a higher risk. These findings suggest that metabolomics data capture additional risk beyond traditional risk factors for AF. Specifically, an AF prediction model based solely on age, sex, AF-PRS, and metabolomics had higher predictive value than models based on the CHARGE-AF score with or without AF-PRS. This is the first study, to our knowledge, to demonstrate the incremental benefit of metabolomics for AF risk prediction for models incorporating both clinical and polygenic risk.

AF Risk Prediction Models

Clinical risk scores have been widely developed in predicting incident AF, with over 20 different models that have been validated in the community. The 2023 ACC/AHA/ACCP/HRS Guidelines support CHARGE-AF as the most widely replicated model, along with the C_2HEST model for Asian cohorts.[23] Newer clinical risk scores include Electronic Health Record-AF, which takes advantage of increased data that are readily available in the electronic health record, and HARMS_2-AF, which incorporates more lifestyle risk factors that are modifiable and patient-centric.[24,25] The best performing clinical risk scores have demonstrated AUC values between 0.7 and 0.8, which is considered fairly good for risk prediction.[26] As knowledge of both family history and genetic variants associated with AF has increased, AF PRS with varying numbers of single-nucleotide polymorphisms have been developed to add to AF risk prediction models. Although performance of AF-PRS as a standalone risk score has been poor, its values exceeds that of the clinical score alone when added to clinical risk factors.[27] Recently, there has been increased interest in exploring metabolomic profiles that correspond to disease states. Buergel et al[16] showed that a neural network trained on a metabolomic panel stratified AF risk quite well, with individuals in the highest decile of metabolomic risk having 8-fold higher odds of having AF than individuals in the lowest decile of metabolomic risk. Other studies that have applied metabolomics to predict incident AF risk have involved small cohorts and limited metabolomic panels. Alonso et al[28] used a multivariable-adjusted nontargeted metabolomics approach in the ARIC to identify bile acids, glycolithocholate sulfate, and glycocholenate sulfate as metabolites associated with HRs of 1.10 to 1.22 for incident AF. Another longitudinal study in the FHS by Ko et al[29] found no metabolites associated with incident AF risk. To our knowledge, no prior studies have assessed incident AF prediction using metabolomics in larger cohorts nor benchmarked the predictive metrics of metabolomics when combined with clinical and genomic risk scores.

Specific Metabolites and AF Risk

The inclusion of specific metabolites in our models may provide mechanistic insights into AF pathogenesis. We found certain metabolites involved in ketone body metabolism (eg, 3-hydroxybutyrate, acetoacetate, and acetone) to be associated with increased AF risk. Bode et al[30] describe the concept of metabolic remodeling in AF, a process in which AF causes a shift in glucose, lipid, and ketone metabolism occurring in the atria. Similarly, ketogenic amino acids like tyrosine and phenylalanine have been linked to oxidative stress and inflammation, processes implicated in AF development.[14] Conversely, we found that metabolites in the lipid classes ranging from high-density lipoprotein, LDL, to VLDL were consistently associated with a lower risk of AF. These findings corroborate prior work that has described a lipid paradox in AF pathogenesis, a phenomenon in which there is an inverse relationship between total cholesterol, LDL, and high-density lipoprotein and the incidence of AF.[31,32] A recent analysis by Chen et al[33] in the UKBB showed an inverse relationship between lipids (more specifically remnant cholesterol) and AF incidence. However, Ouyang et al performed a similar analysis in a USA-based prospective cohort that showed a direct relationship

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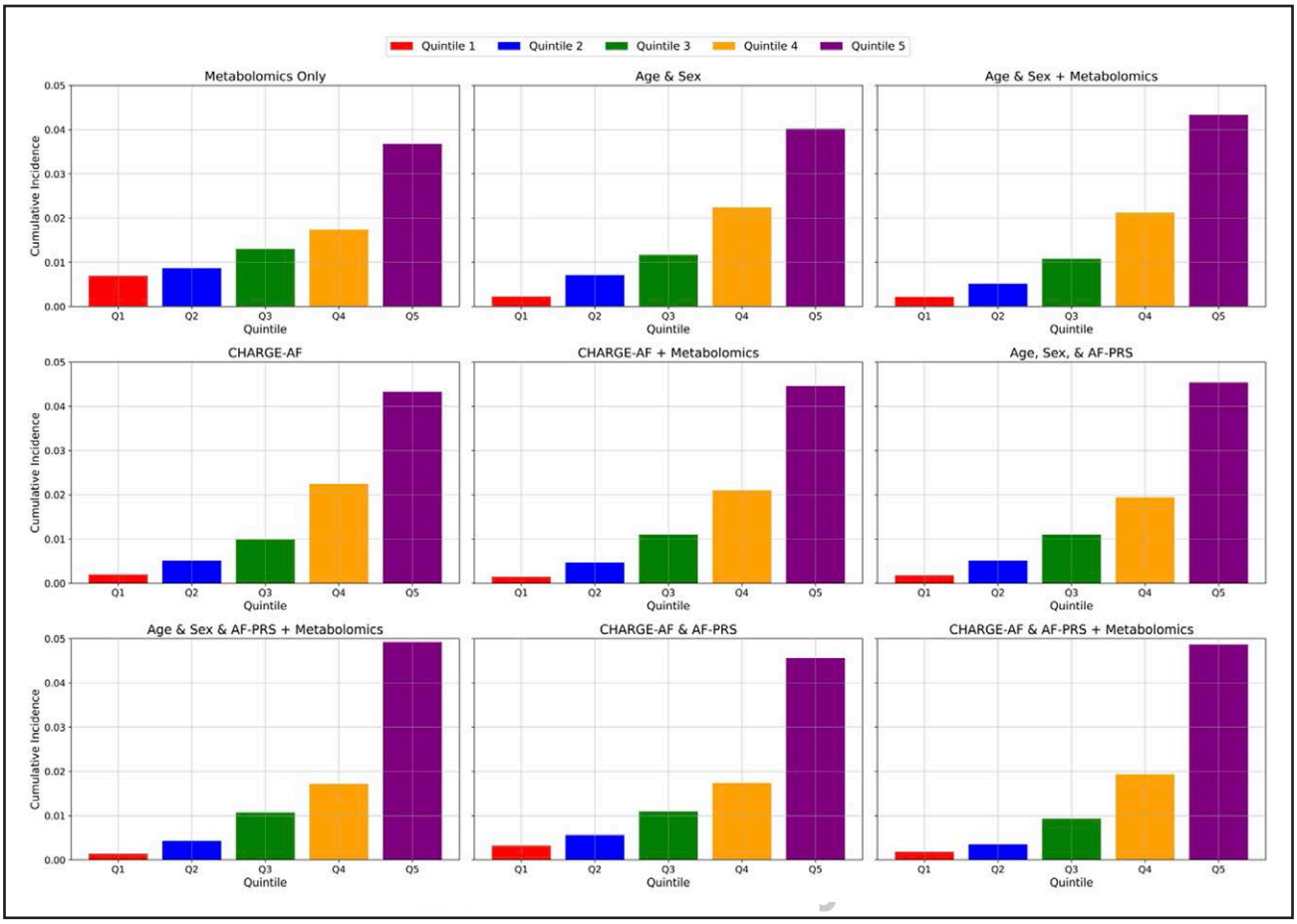

Figure 2. Cumulative incidence of atrial fibrillation (AF) stratified by risk quintiles obtained for each model on the validation set. AF-PRS indicates AF-polygenic risk score; and CHARGE-AF, Cohorts for Heart and Aging Research Genomic Epidemiology–Atrial Fibrillation.

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between remnant cholesterol and AF incidence.[34] The conflicting findings may be due to inherent differences in the United Kingdom and United States patient cohorts, lack of inclusion of unmeasured confounders, and presence of lengthy follow-up times (12.8 and 22.4 years, respectively), which may have violated the proportional hazards assumptions of their models.[35] We had limited the follow-up time to 5 years in our study to minimize the impact of this limitation. We also found that linoleic acid, an essential polyunsaturated fatty acid found mainly in vegetable oils, nuts, and oily seeds, was found to be consistently associated with a lower risk of AF, and such an association has been reported in prior cohort studies.[36] The mechanism behind this association remains unclear. Overall, our study findings suggest that specific metabolic pathways modulate AF risk. Further studies are needed to validate these associations and explore whether measuring specific metabolite levels can lead to actionable clinical decisions.

Clinical Implications

A risk model that involves an individual's age, sex, and serum biomarkers presents an appealing way to predict 5-year AF risk. While there remain significant hurdles to obtaining genomic and metabolomic measurements at scale, there is an opportunity for enhanced risk stratification once accessibility, computational time, and cost improve. Specifically, it has been shown that aggressive lifestyle modifications, including implementing weight loss goals, increasing exercise capacity, and reducing alcohol and tobacco use, can all decrease both the incidence of AF as well as AF burden.[37] Ultimately, an accurate and personalized tool to identify individuals with a high 5-year risk of AF can improve our ability to take advantage of the benefits of such preventive strategies.

Study Limitations

Our study has several limitations. First, while both the CHARGE-AF score and AF-PRS have been externally validated, our current combined model has only been internally validated with the UKBB, which may lead to overestimation of its performance. Further evaluation of our prediction tool using an external validation cohort with comparable metabolomic, genomic, and clinical data is warranted. Furthermore, the UKBB cohort is largely homogenous, lacking racial and ethnic

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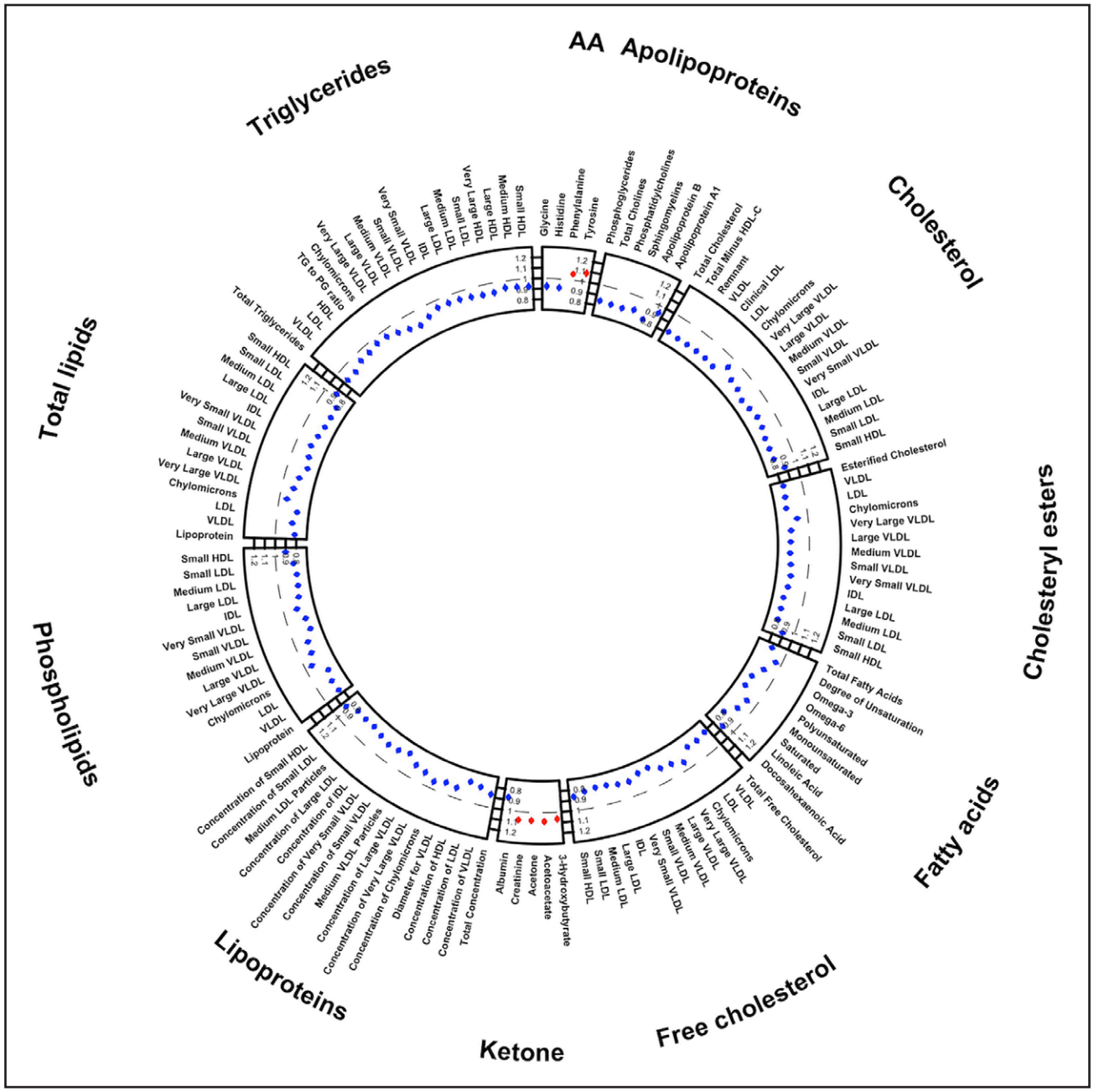

Figure 3. Individual metabolite associations with incident atrial fibrillation (AF).
Circular plot of hazard ratios corresponding to individual metabolites for incident AF after adjustment by age, sex, and AF-polygenic risk score (AF-PRS). Metabolite hazard ratios are grouped by metabolite type in solid black boxes. Metabolites independently associated with AF are bolded. Positive associations are colored red, while negative associations are colored blue. AA indicates amino acids; HDL, high-density lipoprotein; LDL, low-density lipoprotein; and VLDL, very low-density lipoprotein.

diversity, and may not fully capture the heterogeneity of AF pathogenesis in other populations. Secondly, our risk tools are based on Cox proportional hazards models, which do not address competing risks of death and incident AF. Third, circulating biomarkers, such as BNP (B-type natriuretic peptide) and hs-CRP (high-sensitivity C-reactive protein), which are known to be associated with AF risk was not included in our current study analysis as our study focus was on metabolomics. Finally, although the inclusion of metabolomics improved risk prediction, the specific biological mechanisms linking individual metabolites to AF remain unclear. The effect of metabolites such as linoleic acid and cholesterol that were found to modify AF risk may be confounded by dietary patterns and other lifestyle factors that were not included in our clinical variables. Further directions involve incorporating established biomarkers in tandem with metabolomics, validating the metabolomics models in external, diverse cohorts, evaluating how metabolomics can aid in predicting AF

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outcomes like stroke and heart failure, and performing prospective analyses to better characterize the clinical implications of identifying patients with high-risk AF using metabolomics.

Conclusions

The addition of metabolomics to clinical and genomic risk scores for AF risk prediction significantly improves performance in a large population. A risk stratification tool utilizing age, sex, serum metabolomics, and AF-PRS provides excellent 5-year AF risk prediction. We find several individual metabolites that significantly modify AF risk. The specific mechanisms by which these metabolites alter AF risk motivate further exploration.

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Affiliations

Division of Cardiology, Department of Medicine (S.P., J.P., S.B., A.C., S.M.M., B.B.L., J.C.L., J.W.C.), Department of Physiology and Biophysics, Institute for Computational Biomedicine (O.E., J.K.), Caryl and Israel Englander Institute for Precision Medicine (O.E., J.K.), and Cardiovascular Research Institute (J.C.L.), Weill Cornell Medicine, NY.

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Disclosures

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Supplemental Material

Tables S1–S5
Figures S1 and S2

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